



National Highways Authority of India

Ministry of Road Transport & Highways, Government of India



NHAI HACKATHON 7.0

FHELI: SOVEREIGN EDGE BIOMETRICS & LIVENESS SYSTEM

Fully Homomorphic Encryption | Zero-Knowledge Proofs | Edge Neural Networks

Offline-First Architecture | Hardware Agnostic | Open Source

Title: Develop a mobile based secure offline facial recognition and liveness detection system for remote locations.

Objective: To develop a highly accurate, lightweight, and entirely offline facial recognition and liveness detection algorithm that can be seamlessly integrated into the existing Datalake 3.0 app, ensuring uninterrupted operations in zero-network zones.

Team



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Team Members

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Offline Biometric Authentication in Zero-Connectivity Zones

SECURITY VULNERABILITY

Storing raw biometric templates in plaintext exposes them to extraction if the device is rooted or physically compromised. Standard offline mobile biometrics lack encryption-at-rest for face vectors.

LIVENESS VULNERABILITY

Simple RGB cameras are fooled by high-res color prints, tablet playbacks, twins, or paper cutouts where a proxy places their eyes behind a photo. No reliable 3D depth verification exists.

HARDWARE CONSTRAINTS

Budget-to-mid-range field devices have ≤ 3 GB RAM, weak NPUs, and limited battery. The biometric system must run in sub-second times and fit within a ≤ 20 MB bundle size.

THE FHELI SOLUTION

8.45 MB Bundle

Channel Pruning + INT8 PTQ compresses BlazeFace + FaceMesh + MobileFaceNet pipeline from 37.2 MB to 8.45 MB

CKKS Homomorphic Matching

Cheon-Kim-Kim-Song scheme compares face vectors in encrypted ciphertext. Raw templates are never decrypted in memory.

Passive 3D Liveness

Depth contours, corneal specular reflection, pupil nystagmus FFT, and 3D motion parallax flow verification

zk-SNARK Decentralized Proofs

256-byte Groth16 proofs verify shift times and site boundaries offline, syncing when connectivity is restored

8.45 MB Bundle Size

<108 ms Pipeline Latency

~2 ms FHE Matching

>95% Face Accuracy

Global Edge Biometric Landscape Comparison

System	Primary Mechanism	Core Limitations
US (Apple FaceID)	Structured Light IR Dot Projection	Requires custom TrueDepth camera hardware; matches templates in plaintext in RAM
China (SenseTime)	Deep CNNs on NPUs	Large footprint (>50MB), high power drain; centralized tracking of GPS/biometrics
Israel (Oosto)	Local database matching	Caches plaintext templates in SQLite; high-cost proprietary hardware terminals
Fheli (Ours)	Agnostic RGB + FHE + zk-SNARKs	Zero hardware dependencies; homomorphic ciphertext matching; decentralized, zero GPS leak

KEY DIFFERENTIATORS

Hardware Independence Operates on standard RGB cameras using passive specular calculations. No IR dot projectors required.	RAM Extraction Defense Vectors matched in ciphertext state. A memory dump yields only random cryptographic noise.	Metadata Leakage Prevention zk-SNARKs prove shift validity without sending coordinates or ID strings to the cloud.
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0	256 bytes	100%	0
Hardware Dependencies	Proof Size	Open Source Stack	Cloud GPS Leak

Five Integrated Technical Pillars

01

FHE VECTOR MATCHING

CKKS homomorphic scheme executes 128-D dot product inside ciphertext space. Raw biometric templates never decrypted.

~2 ms matching latency

$N=2^{11}$, $\Delta=2^{30}$

02

CORNEAL SPECULAR CHECK

Mobile display changes colors at 100ms intervals. Camera captures specular reflections on cornea to verify 3D curvature.

Sunlight fallback mode

EER + MAR at >80,000 lux

03

3D PARALLAX FLOW

Tracks nose-to-cheek optical flow from saccadic tremors. Real 3D faces produce depth parallax; flat photos yield zero.

Delta > 0.0003 threshold

Real vs spoof detection

04

zk-SNARK ATTENDANCE

Groth16 generates 256-byte proofs on-device. Verifies match score, GPS boundary, and shift window without revealing data.

<108 ms UI response

Async background proof gen

05

NEUROMORPHIC SLICING

Native C++ frame slicing monitors temporal motion deltas. TFLite models execute only when motion threshold is crossed.

80% idle CPU reduction

Prevents overheating

$$\text{Similarity}(A,B) = \text{Sum}(A_i * B_i) \mid \text{CKKS: } C_{\text{match}} = \text{Sum}(C_{\text{enroll},i} (x) C_{\text{live},i}) \mid \text{Decrypt}(C_{\text{match}},sk) \geq 0.85 = \text{PASS}$$

37.2 MB Compressed to 8.45 MB at Sub-Second Speed



Component	Architecture	Precision	Original	Quantized	Latency
Face Detection	BlazeFace	INT8	1.2 MB	150 KB	~5 ms
Landmarks	FaceMesh	INT8	12.0 MB	2.5 MB	~18 ms
Recognition	MobileFaceNet	INT8	24.0 MB	5.8 MB	~85 ms
Operator Fusion	Conv+BN+ReLU	INT8	-	-	~0 ms
TOTAL	3-Stage Pipeline	INT8	37.2 MB	8.45 MB	~108 ms

KEY METRICS

77.3%

Size Reduction Achieved

4.4x

Compression Ratio

<108 ms

End-to-End Pipeline Latency

DEMOGRAPHIC ADAPTABILITY

Indian Demographics Fine-Tuning

MobileFaceNet fine-tuned on IITK Face Database + VGGFace2 subset with Indian faces across Fitzpatrick Scale Types III to VI, diverse hairstyles, and facial features.

Target accuracy: >95% reliably maintained.

Low-Light / Shadow Preprocessing

Native JSI pre-processing wrapper automatically runs CLAHE (Contrast Limited Adaptive Histogram Equalization) to normalize contrast across face regions before TFLite inference.

Handles harsh shadows & low-light.

Security Architecture & Offline Sync Protocol

THREAT MITIGATION (OWASP MOBILE TOP 10)

- Physical DB Extraction** — SQLCipher AES-256 encryption prevents decryption even if database files are copied from rooted devices.
- Replay & Spoof Attacks** — Active blink/smile verification combined with passive 3D depth variance blocks all 2D prints and screen playbacks.
- Attendance Manipulation** — SHA-256 signature binds RecordID + Timestamp + GPS + MatchScore. Any tampering voids the hash.
- Key Stealing** — Keys generated inside hardware-backed Secure Enclave / KeyStore. Unextractable even with root privilege.
- Network Interception** — AWS Lambda upload batches signed with HMAC-SHA256 protecting against MITM payload alterations.

OFFLINE SYNC & PURGE PROTOCOL

- 1. NetInfo Monitor**
Listens for connection changes. Detects when device comes back online.
- 2. Batch Generation**
Unsynced records aggregated into signed JSON with deviceId, records array, and HMAC-SHA256 batchHash.
- 3. AWS Gateway Upload**
Signed payload transmitted to AWS endpoint over secure channel.
- 4. Purge-on-Receipt**
On successful response signature, local records immediately deleted to prevent storage bloat.

PROJECT DIRECTORY STRUCTURE		
android/app/src/main/jni/ C++ JSI compilation targets	src/components/ CameraView.tsx, LivenessOverlay.tsx	src/utils/liveness.ts EAR, MAR, Parallax checks
ios/FheliEngineFramework/ Swift JSI Native Bridges	src/database/ DBManager.ts (SQLCipher)	src/utils/ZKProver.ts snarkjs Groth16 prover wrapper

Academic Foundation & Open-Source Compliance

CRYPTOGRAPHY & PRIVACY

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FACIAL MODELS & LIVENESS DETECTION

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OPEN-SOURCE TECHNOLOGY STACK

TensorFlow Lite
Apache License 2.0

OpenFHE
BSD 2-Clause License

snarkjs
MIT License

MediaPipe
Apache License 2.0

SQLCipher
BSD 3-Clause License

Fheli is built exclusively on open-source technologies requiring no additional licensing fees.
All cryptographic operations use formally verified academic schemes with peer-reviewed security proofs.